MIL-N-28935B
5 October 1982
SUPERSEDING
MIL-N-28935A
11 December 1975

MILITARY SPECIFICATION

NET, CARGO, AERIAL DELIVERY

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers requirements for one type and size cargo net fabricated of synthetic rope.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and Standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-B-585	- Box; Wood, Wirebound
PPP-B-601	- Box; Wood, Cleated-Plywood
PPP-B-621	- Box; Wood, Nailed and Lock-Corner

MILITARY

MIL-R-17343	- Rope, Nylon
MIL-R-24049	- Rope, Polypropylene
MIL-R-30500	- Polyester

Beneficial comments (recommendations, additions, deletions), and any pertinent data which may be of use in improving this document, should be addressed to Commandant of the Marine Corps, Headquarters Marine Corps (Code LMA-1), Washington, DC 20380, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426), appearing at the end of this document, or by letter.

FSC 1670

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STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Table for Inspection by Attributes
MIL-STD-129	- Marking for Shipment and Storage
MIL-STD-171	- Finishing of Metal and Wood Surfaces
MIL-STD-1188	- Commercial Packaging of Supplies and Equipment

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

*2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DODISS and the supplement thereto, if applicable.

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ANSI/ASTM A668-79a - Steel Forgings, Carbon and Alloy, for General Industrial Use.

(Copies may be obtained from American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103. Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 First article. Unless otherwise specified (see 6.2), as soon as practicable after award of the contract or order, and prior to starting regular production, the manufacturer shall submit a completely assembled cargo net for inspection and approval by the contracting officer or his authorized representative. The first article shall be representative of the construction, workmanship, components and materials to be used during production. Approval of the first article does not preclude the requirements for performing the quality conformance inspection.

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*3.2 Materials. Materials used shall be free from defects which would adversely affect the performance or maintainability of individual components or of the overall assembly. Materials not specified herein shall be of the same quality used for the intended purpose in commercial practice. Unless otherwise specified herein, all equipment, material, and articles incorporated in the work covered by this specification are to be new and fabricated using materials produced from recovered materials to the maximum extent possible without jeopardizing the intended end use of the item.

3.2.1 Rope. The rope shall conform to MIL-R-17343, MIL-R-30500 or type I of MIL-R-24049 except that the thread shall be approximately 15,000 denier.

*3.2.2 Metal. Fittings shall meet or exceed the requirements of ANSI/ASTM A 668-79a, and shall have a 4 to 1 safety factor.

3.3 Design and construction. The design and construction of the net shall conform to the following requirements.

*3.3.1 Net mesh. The mesh portion of the net shall be formed of 4,000-pound (1814 kg) minimum working strength rope specified in 3.2.1. The rope shall form a mesh of 7 $\pm$0.5 inch (177.8 $\pm$ 12.7 mm) centers. The method of joining the intersecting lines of rope must exceed the tensile strength of the rope used to fabricate the net; proof shall be in the form of a contractor's certificate or an actual test. The meshes shall be joined with a 1-inch (25.4 mm) diameter rope to serve as a boundry line. The center of the net shall be indicated.

*3.3.2 Attachment fittings. The net shall have rings and snap hooks of material specified in 3.2.2. The four corners of the net shall be provided with oval steel rings approximately 3 X 6 inches (76.2 X 152.4 mm), having a working capacity of 5,000 pounds (2268 kg). Two adjacent sides of the net shall be provided with 5,000-pound (2268 kg) working capacity snap hooks midway between the corner rings. The remaining two sides shall have a 5,000-pound (2268 kg) working capacity pear-shaped or round steel ring installed midway between the corner rings.

*3.4 Performance. The net shall have a rated lifting capacity of 20,000 pounds (9072 kg), with a safety factor of 4 to 1, and shall be capable of passing the safety factor proof load specified in Section 4.

*3.5 Dimensions. The net shall measure 20 X 20 feet (6.1 X 6.1 meters). Such dimensions are for the basic net structure and do not include rings and the fabrication needed to attach rings to the net. Dimensions and tolerances not specified herein shall be as close as is consistent with best shop practices.

3.6 Finish and color. All hardware shall be hot dipped galvanized finish No. 1.9.4.1 or electrodeposited finish No. 1.9.1.2 in accordance with MIL-STD-171. There are no color requirements for the rope except that fluorescent or phosphorescent coloring agents shall not be used.

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3.7 Workmanship. The net, including all parts and accessories, shall be fabricated and finished in a workmanship quality of standard commercial practice. Rope shall be sound and free from severed or torn yarns or strands. Metal shall be free from burrs, sharp edges or other defects.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Certificate of compliance. When certificates of compliance are submitted as required by the contract or purchase order, the Government reserves the right to check test such items to determine the validity of certification.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3)
b. Quality conformance inspection (see 4.4)
c. Packaging inspection (see 4.7)

4.3 First article inspection. The first article shall be subjected to the visual examination in 4.5 and the tests in 4.6.

4.4 Quality conformance inspection. The visual examinations and tests comprising quality conformance inspection are listed in Table I.

Table I - Examination and Tests.

Category	Requirement	Test Method
Visual examination	Section 3	4.5
Tests:		
Safety factor	3.4	4.6.1
Proof load	3.4	4.6.2

4.4.1 Sampling.

4.4.1.1 Inspection lot. A lot shall consist of all nets made of the same materials, components and essentially under the same conditions and offered for acceptance and delivery at one time.

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4.4.1.2 Visual examination and tests. For examination and tests, a random sample of nets shall be selected from each lot in accordance with MIL-STD-105, inspection level S-2. The acceptable quality level (AQL) shall be 2.5 defects per hundred units (DHU).

4.5 Visual examination. The net shall be visually examined for the major defects listed in Table II.

Table II - Visual Examination.

Examine	Defect
Material	Material and components not as specified
	Any component missing, broken malformed or damaged
Construction	Not as specified
	Any cut or damage affecting serviceability
	Dimensions not as specified
Workmanship	Not as specified

*4.6 Tests. Tests shall be as follows:

*4.6.1 Safety factor. The net shall be tested for the 4 to 1 safety factor by lifting a load of 80,000 pounds (36,287 kg). The load shall be lifted clear of the ground and held for 30 seconds without any failure of net mesh or hardware components.

*4.6.2 Proof load. The proof load test of the net shall be conducted by lifting a 20,000 pound (9072 kg) load. The load shall be held free of the ground for 2-1/2 hours, with no damage to the net or permanent visual deformation of any hardware. After completion of test, measure outer dimensions. If net has more than 10% stretch, lay flat for 24 hours and remeasure. If after 24 hours the outer dimensions show a stretch in excess of 10%, this shall be cause for rejection.

*4.6.3 Drop. The net shall be drop tested by lifting a 20,000-pound (9072 kg) load. The load shall be elevated to a height of 8 feet (2.4 meters) and dropped to bare earth. Repeat 25 times. The net shall retain the load.

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4.7 Packaging. A visual examination shall be made to determine compliance with packaging requirements of Section 5. Defects shall be scored in accordance with the list below. The sample unit shall be one shipping container fully prepared for delivery. The inspection level shall be S-2 and the AQL shall be 4.0 DHU.

Examine	Defects
Materials	Any component missing or damaged
Workmanship	Incomplete closure, bulging or distortion of containers
Markings	Incorrect, illegible, of improper size, location, sequence or method of application

5. PACKAGING

5.1 Packaging. No packaging required.

5.2 Packing. Packing shall be level A, B, or commercial as specified (see 6.2).

*5.2.1 Level A. Each cargo net shall be folded or bundled to the smallest practical dimension and packed within a close fitting box conforming to PPP-B-585, class 3, PPP-B-601, overseas type or PPP-B-621, class 2. Closure and strapping shall be in accordance with the appendix of the applicable box specification.

5.2.2 Level B. Each cargo net shall be packed as specified in 5.2.1 except that the shipping containers shall be domestic type or class 1 as applicable.

*5.2.3 Commercial packing. Each cargo net shall be packed in a manner to insure carrier acceptance and safe delivery to destination at the lowest transportation rate in accordance with MIL-STD-1188.

5.3 Marking. In addition to any marking required by the contract or order, shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The net is intended for use in transporting external cargo by helicopter in aerial delivery operations.

*6.2 Ordering data.

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*6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification.
b. Whether first article inspection is required (see 3.1).
c. Level of packing required (see 5.2).

*6.2.2 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements Lists (CDRL) the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of DAR 7-104.9 (n) (2) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraphs.

Paragraph No.	Data Requirements	Applicable DID No.	Options
4.3	First Article Inspection Procedure	DI-T-4901	----
4.3	First Article Inspection Report	DI-T-4902	----
4.4	Production/Acceptance Inspection Procedures	DI-T-4903	----
4.4	Production Inspection Reports	DI-T-4904	

(Copies of DID's required by contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.3 Instruction for first article. When first article is required, the contracting officer shall specify the location to which the first article is to be furnished for examination and test.

*6.4 Marginal notations. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship of the last previous issue.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
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DOCUMENT IDENTIFIER AND TITLE

MIL-N-28935B NET, CARGO, AERIAL DELIVERY

NAME OF ORGANIZATION AND ADDRESS	CONTRACT NUMBER
	MATERIAL PROCURED UNDER A ☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

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